

Trainings ▾

Pin Replacement

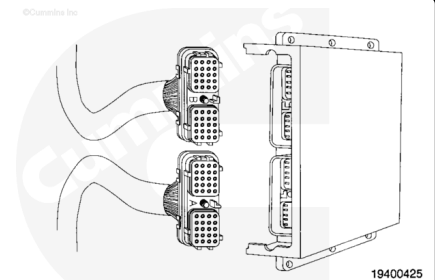
Deutsch

The 40-pin Deutsch connector is used to attach the OEM harness to the ECM and it is also used on the engine harness.

This connector is also used on the engine harness adaptor cable and engine harness extension cables, which allow the ECM and the engine harness to be connected over varying distances.

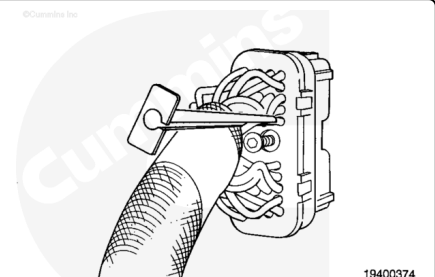
The engine harness adaptor cable is 5 ft in length.

The engine harness extension cables come in length's of 5 ft, 10 ft, and 20 ft.

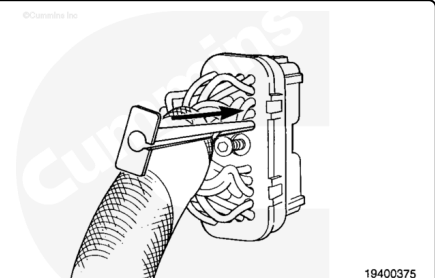


Use Deutsch extraction tool, Part No. 3824815, over the wire to remove a pin from the connector.

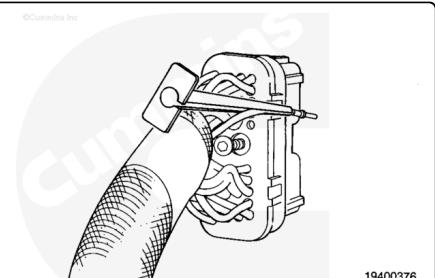
Note : Replace one wire at a time. If more than one wire needs replacement, attach a lettered tag to each wire removed. Refer to the wiring diagram in Section E of this manual.



Push the tool into the connector about 25 mm [1 inch] until it bottoms on the terminal flange.



Hold the tool on the terminal flange and pull the wire and the connecting pin out of the connector. Note and record the hole from which the pin is removed.



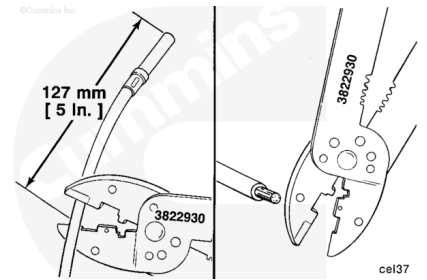
Note : The repair wire is 127 mm [5 inches] long.

The female terminal is Part No. 3824810.

The male terminal is Part No. 38224809.

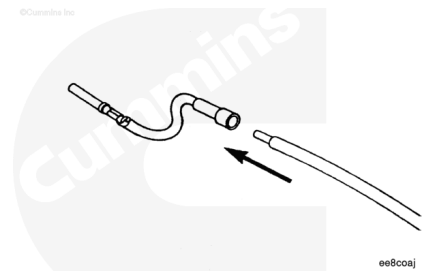
Use wire crimp tool, Part No. 3822930, to cut 127 mm [5 inches] off the wire and pin.

Use the crimp tool to remove 6 mm [1/4 inch] of insulation from the wire.

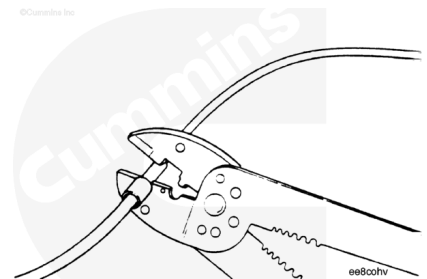


Install the correct repair wire on the bare wire.

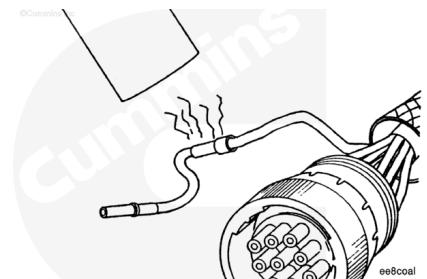
Note : Make sure the bare wire extends into the splice connector.



Use the wire crimp tool to crimp the repair wire onto the bare wire.



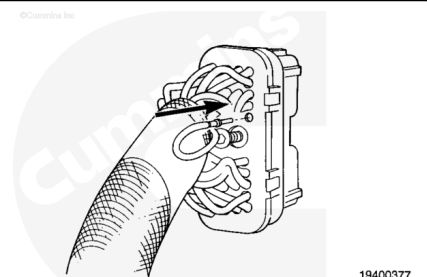
Use a heat gun, Part No. 3822860, or open flame to heat the shrink tubing around the wire. The tubing will shrink and make the connector waterproof.



Insert the pin into the correct hole of the connector.

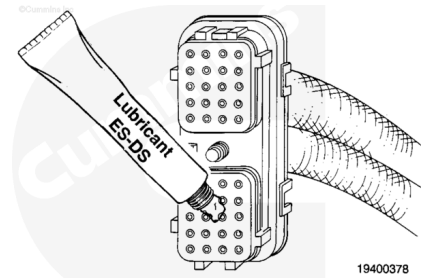
The pin **must** click into place and hold the wire in the connector.

Pull the wire gently to make sure it is seated in the connector.



⚠ CAUTION ⚠

Use only Cummins recommended lubricant DS-ES, Part No. 3822934. Other lubricants, such as lubricating oil or grease, in the connectors can cause ECM damage, poor engine performance, or premature connector pin wear.



Apply a small amount of lubricant to the connector terminals.
Do **not** fill the entire connector cavity with lubricant.

Connector Replacement

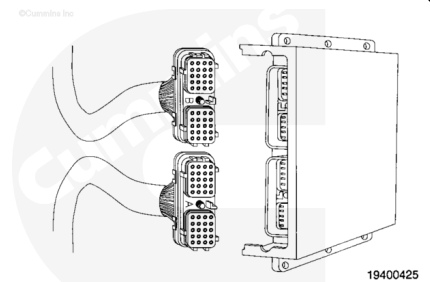
Deutsch

The 40-pin Deutsch connector is used to attach the OEM harness to the ECM and it is used on the engine harness.

This connector is also used on the engine harness adaptor cable and engine harness extension cables, which allow the ECM and the engine harness to be connected over varying distances.

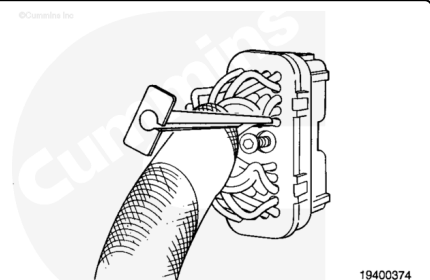
The engine harness adaptor cable is 5 ft in length.

The engine harness extension cables come in length's of 5 ft, 10 ft, and 20 ft.

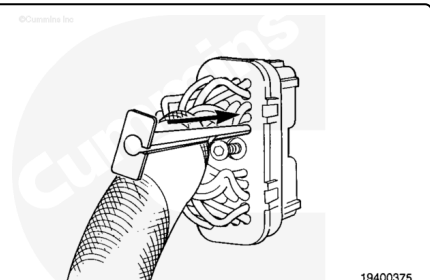


To replace the connector, use Deutsch extraction tool, Part No. 3824815, over each wire to remove all pins from the connector.

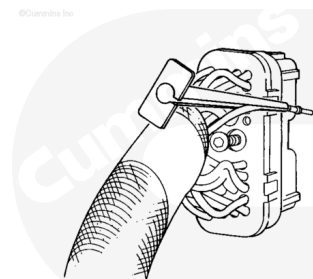
Note : Replace one wire at a time. Attach a lettered tag to each wire removed. Refer to the wiring diagram in Section E of this manual.



Push the tool into the connector about 25 mm [1 inch] until it bottoms on the terminal flange.



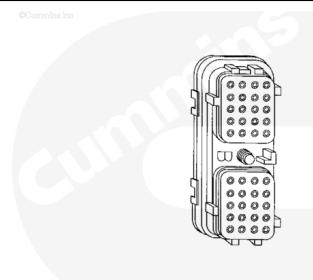
Hold the tool on the terminal flange and pull the wire and the connecting pin out of the connector. Note and record the hole from which the pin is removed.



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The replacement OEM harness connector is Part No. 3824818.

The replacement engine harness connector is Part No. 3824817.

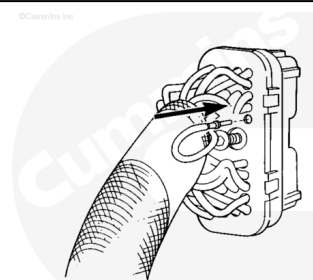


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Insert the pins into the correct holes of the replacement connector.

Each pin **must** click into place and hold the wires in the connector.

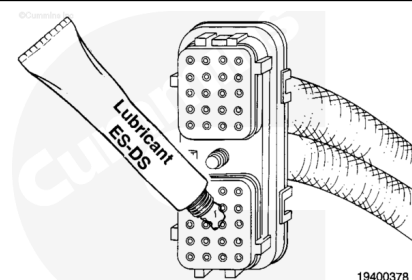
Pull each wire gently to make sure it is seated in the connector.



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⚠ CAUTION ⚠

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Apply a small amount of lubricant to the connector terminals. Do **not** fill the entire connector cavity with lubricant.